JavaScript Functions · supersimple.dev · youtu.be/EerdGm-ehJQ (3:34:12 – 4:14:34)

Function Basics

Functions save a block of code to be reused. Code inside { } does **not** *run until called.*

```
function sayHello() {
  console.log('Hello');
}
```

Creates a reusable, named block of code.

Function Syntax

Syntax = rules for writing JavaScript functions.

```
function  sayHello()  { ... }
```

Keyword	Name	Body
(creates it)	*(labels it)*	*(code runs here)*

Calling a Function

```
sayHello();
```

Runs the body. Can be called as many times as needed.

Parameters & Arguments

Pass data **into** *a function. Parameters = placeholders. Arguments = actual values.*

```
function calcTax(cost) { }
```

cost *is a* **parameter** *— temporary variable inside the function.*

```
calcTax(2000);
```

2000 *is the* **argument** *— value passed in, caught by* **cost**.

```
function greet(name, age) { }
```

Multiple params → separate with commas.

```
greet('John', 25);
```

Args match params **in order** *→ name='John', age=25.*

```
function greet(name = 'Hi') { }
```

Default param *— used when no argument is passed.*

Return Statement

Gets a value **out** *of a function.* **Stops execution immediately** *when hit.*

```
return 5;
```

Outputs **5** *from the function and stops immediately.*

`return mathCalculation;`	*Outputs the value stored in a variable.*
`let tax = calcTax();`	*Captures the returned value into a variable.*
`return;`	*Returns* **undefined** *— exits early (e.g. inside an if-check).*

Variable Scope

Scope = where a variable can be accessed.

Local Scope	*Variable inside* **{ }** *— only accessible within those braces.*
Global Scope	*Variable outside* **{ }** *— accessible anywhere in the file.*
Naming Conflict	*Same name is fine* **only if** *they are in two different local scopes.*

Hoisting

JS reads the whole file first and moves function declarations to the top before running.

`playGame();  // works!`	*Called* **before** *it is defined — still works.*
`function playGame() { ... }`	*⚠ Only standard functions hoist — not arrow functions.*